

Step 3: Flowchart for the Pet Feeder System

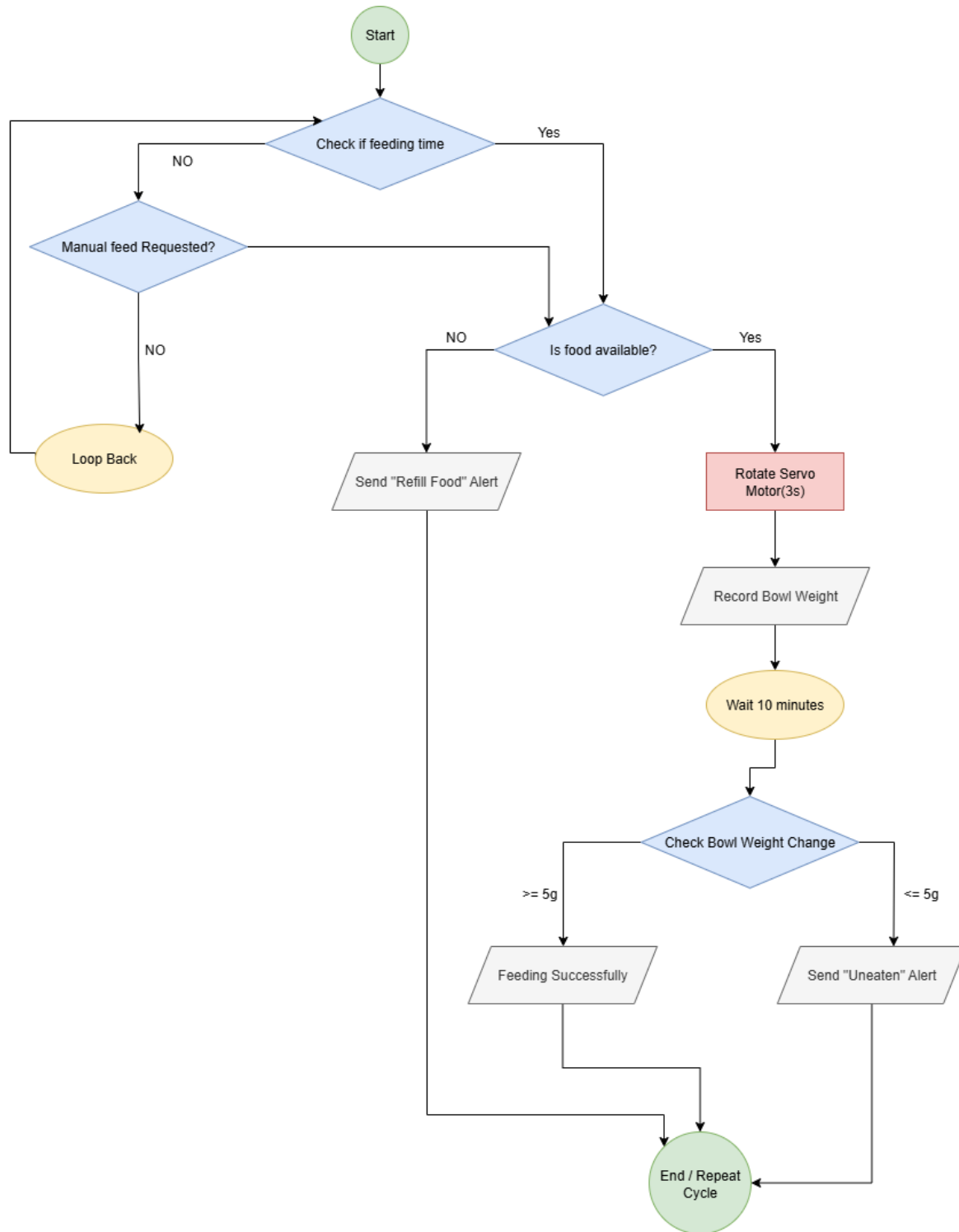


Figure 2: System Flowchart

Logic Behind the Flowchart:

Start → system initializes.

1. Check Feeding Time

- If current time = 8:00 AM or 6:00 PM → proceed.
- Else, check if a manual feed request is made.

2. Manual Feed Request

- If pressed → proceed to food availability check.
- Else → return to the waiting state.

3. Check Food Availability

- If food is available in the storage → proceed to servo motor action.
- If not available → send “Refill Food” Alert and return to waiting.

4. Dispense Food

- Rotate servo motor for 3 seconds to release food.
- Record bowl weight before feeding.

5. Wait and Monitor Bowl

- Wait 10 minutes.
- Compare bowl weight before and after.

6. Check Bowl Weight Change

- If weight decreased by $\geq 5g$ → feeding is successful.
- If weight decreased by $\leq 5g$ → send “Uneaten Food” Alert.

7. Return to Waiting State

- Repeat cycle until next feeding time or manual request